

Rational-Exponential Iterated Integrals and Quantum Polylogarithms

Absolute fixed-parameter completeness and a pentagon crossed product

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Abstract

We study Goncharov's quantum polylogarithms at a fixed positive irrational parameter. A Laplace transform of the cumulative momentum denominators gives an ordered iterated-integral presentation, with the upper $i0$ prescription justified on positively shifted contours. We compute the constant-linear space of translated weight-zero Fourier letters as a two-variable localization modulo Laurent polynomials. A residue argument makes this alphabet independent modulo meromorphically exact differentials, yielding shuffle completeness at all positive integral weights and depths in one common translation variable. We also present the entire weight-zero differential field. After adjoining e^x and $e^{x/\hbar}$, it is a rational function field in the derivatives of explicit normalized mixed towers. These derivatives are algebraically independent: any dependence would produce an additive difference telescoper, which a single-pole-orbit obstruction excludes. Combining the two presentations classifies nonlinear coefficient relations and positive-weight relations over $\mathbb{C}(x)$, without an unspecified meromorphic coefficient field. A separate Ore-denominator argument proves the faithful cyclic crossed-product presentation of Goncharov's pentagon operator algebra. The scalar and operator statements remain separate. No parameter-changing completeness, scalar quantum-cluster action, or arithmetic specialization theorem is asserted.

Keywords. Quantum polylogarithms; rational-exponential integrals; iterated integrals; differential algebraic independence; difference equations; shuffle algebra; quantum dilogarithm; Ore domain.

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1 Introduction and principal statements

Goncharov defines quantum polylogarithms by multiple Fourier integrals with cumulative momentum denominators and proves modular difference equations, continuation properties, and shuffle identities [4]. Their integrands are rational expressions in the momentum variables and their exponentials. We use his term *rational-exponential integrals* for this analytic class. Our first purpose is to exhibit a direct ordered-integral presentation of the positive-integral-weight sector. Our second is to determine the constant-linear letter relations, the nonlinear algebraic structure of the weight-zero differential field, and the polynomial relations among the resulting iterated integrals.

Throughout the completeness theorems, $\hbar$ and finitely many offsets are fixed, and the variable is a common additive translation of the arguments. The shuffle theorem is first proved over the actual single-valued meromorphic weight-zero field. An independent difference-algebra argument then presents that field as a rational function field in an explicit differential transcendence basis. The resulting combined presentation is *absolute* in the sense that its coefficient field over $\mathbb{C}(x)$ is fully presented. It is a functional relation theorem, not a numerical Kontsevich–Zagier theorem, and uses no arithmetic specialization or E -function value lifting.

The analytic reduction is related to Chen’s iterated integrals [2] and to the modified reduced-bar formalism used by Brown [1, Section 3]. The general shuffle machinery is not new. The specific inputs here are the contour-level reduction, the quantum-letter quotient, the residue obstruction for iterated integrals, and a normalized tower presentation of the nonlinear coefficient field. The latter uses the classical connection between additive difference equations, telescoping, and algebraic independence [6, 5]; the needed obstruction is proved directly for these towers. In a separate appendix we give an operator result for the pentagon structure of [3]. It neither uses nor implies an action on the scalar iterated-integral algebra.

1.1 Concurrent and related developments

Other q -deformations of iterated integrals address related but different questions. Hirose introduced iterated q -integrals on the four-punctured projective line with position-dependent shifts and conjectured a word-duality identity [13]; Seki subsequently proved that conjecture using a terminating basic-hypergeometric connector [15]. Blümlein–Gavrilik–Mykhailiv–Schneider construct q -extensions of iterated-integral alphabets and nested sums motivated by quantum field theory [16]. We cite these as distinct deformation theories; we neither identify their functions with Goncharov’s Fourier integrals nor use their word identities to obtain the completeness theorem here.

The contemporaneous double-sine constructions of cyclic Haagerup–Izumi categories by Huang [9] and Gannon–Schopieray–Yadav [10] belong to the finite categorical side of this subject. They provide important context for the arithmetic direction, not inputs to the fixed-parameter residue, coefficient-field, or shuffle proofs below. The relation to RW and the Stark/SIC literature is specified in the arithmetic-motivation subsection.

1.2 The functions and the common translation variable

Use the normalization

$$\operatorname{sh} z = e^z - e^{-z} = 2 \sinh z, \quad H_{a,b}(p) = \frac{1}{\operatorname{sh}(\pi p)^a \operatorname{sh}(\pi \hbar p)^b},$$

where $a, b \in \mathbb{Z}_{\geq 0}$ and $a + b > 0$. The weight-zero Fourier letter is

$$L_{a,b}(z) = -i \int_{\mathbb{R}+i0} e^{-ipz} H_{a,b}(p) dp. \quad (1.1)$$

Dependence on $\hbar$ is suppressed. For $\mathbf{n} = (n_1, \dots, n_m)$ with $n_j \geq 1$, put $|\mathbf{n}| = \sum n_j$ and

$$\mathcal{F}_{\mathbf{a}, \mathbf{b}, \mathbf{n}}^{\hbar}(\omega) = i^{|\mathbf{n}|-m} \int_{(\mathbb{R}+i0)^m} \prod_{j=1}^m \frac{e^{-ip_j \omega_j} H_{a_j, b_j}(p_j)}{(p_1 + \dots + p_j)^{n_j}} dp_1 \dots dp_m. \quad (1.2)$$

This is [4, Definition 1.2], restricted to positive integral weights and nontrivial kernels. The precise contour convention is given in Section 2.

Fix a nonempty finite set $\Delta \subset \mathbb{C}$. Write

$$\omega_j = x + \delta_j - i\pi(r_j + s_j \hbar), \quad \delta_j \in \Delta, \quad r_j, s_j \in \mathbb{Z}. \quad (1.3)$$

Set

$$\Lambda_{\hbar} = i\pi(\mathbb{Z} + \hbar\mathbb{Z}), \quad \Delta/\Lambda_{\hbar} = \{\text{cosets represented by elements of } \Delta\}.$$

Let $\mathcal{W}_{\hbar, \Delta}$ be the complex vector space of actual functions spanned by

$$x \mapsto L_{a,b}(x + \delta - i\pi(r + s\hbar)). \quad (1.4)$$

Let $K_{\hbar, \Delta}$ be the differential subfield of $\text{Mer}(\mathbb{C})$ generated over $\mathbb{C}(x)$ by these functions. Equivalently it is generated as a field by the functions in (1.4) and all their derivatives. Its constants are $\mathbb{C}$.

Define

$$\begin{aligned} \mathcal{R} &= \mathbb{C}[X^{\pm 1}, Y^{\pm 1}], & d_X &= X^{-1} - X, & d_Y &= Y^{-1} - Y, \\ \mathcal{S} &= \mathcal{R}[d_X^{-1}, d_Y^{-1}], & \mathcal{V}_{\hbar, \Delta} &= \bigoplus_{\delta \in \Delta/\Lambda_{\hbar}} \mathcal{S}/\mathcal{R}. \end{aligned} \quad (1.5)$$

A representative δ is chosen in each coset. Different choices identify the summands by multiplication by Laurent monomials.

Theorem 1.1 (Letter presentation and residue exactness). *Suppose $\hbar > 0$ is irrational. There is an isomorphism of complex vector spaces*

$$\mathcal{V}_{\hbar, \Delta} \xrightarrow{\sim} \mathcal{W}_{\hbar, \Delta}, \quad \left[X^r Y^s d_X^{-a} d_Y^{-b} \right]_{\delta} \mapsto L_{a,b}(x + \delta - i\pi(r + s\hbar)). \quad (1.6)$$

The only constant-linear relations in one coset are the two centered modular lowering relations, with the terminal symbol $e_{0,0}$ set equal to zero. Moreover,

$$\Omega \in \mathcal{W}_{\hbar, \Delta} dx, \quad \Omega = dG, \quad G \in \text{Mer}(\mathbb{C}) \implies \Omega = 0. \quad (1.7)$$

In particular $\mathcal{W}_{\hbar, \Delta} dx$ embeds into $K_{\hbar, \Delta} dx/dK_{\hbar, \Delta}$.

For $f_1, \dots, f_m \in \mathcal{W}_{\hbar, \Delta}$ define the ascending Chen integral

$$I[f_1 | \dots | f_m](x) = \int_{-\infty < t_1 < \dots < t_m < x} \prod_{j=1}^m f_j(t_j) dt_1 \dots dt_m. \quad (1.8)$$

Initially all variables lie on the horizontal ray ending at x , in a sufficiently far left half-plane. These integrals converge and give holomorphic functions there. The empty word has value 1.

Theorem 1.2 (Shuffle completeness over the weight-zero field). *Suppose $\hbar > 0$ is irrational and $\Delta \subset \mathbb{C}$ is finite and nonempty. Let $\mathcal{Q}_{\hbar,\Delta}^{>0}$ be the $K_{\hbar,\Delta}$ -algebra generated by all functions (1.2) on the common-translation lines (1.3), with arbitrary depth, $a_j, b_j \geq 0$, $a_j + b_j > 0$, and $n_j \geq 1$. Use the branch continued from the defining strips into $\Re \omega_j < 0$. Then iterated integration induces an isomorphism*

$$K_{\hbar,\Delta} \otimes_{\mathbb{C}} \text{Sh}(\mathcal{V}_{\hbar,\Delta}) \xrightarrow{\sim} \mathcal{Q}_{\hbar,\Delta}^{>0}. \quad (1.9)$$

The isomorphism uses (1.6) on letters. In particular, after choosing an ordered basis of $\mathcal{V}_{\hbar,\Delta}$, the iterated integrals of Lyndon words are algebraically independent over $K_{\hbar,\Delta}$ and generate the algebra. No generic-position hypothesis on the offsets is required: coincident lattice cosets are combined as in (1.5).

Here $\text{Sh}(V) = \bigoplus_{n \geq 0} V^{\otimes n}$ denotes the shuffle algebra, not the tensor algebra with concatenation product. The theorem presents the *iterated-integral extension* of the coefficient field. It does not identify $\mathcal{V}_{\hbar,\Delta}$ with the entire de Rham quotient of that field.

1.3 The coefficient field and the absolute presentation

Choose a representative δ for each coset in $\Delta/\Lambda_{\hbar}$, and let

$$\mathcal{R}_{\Delta} = \{\delta + \pi i(e + f\hbar) : \delta \in \Delta/\Lambda_{\hbar}, e, f \in \{0, 1\}\}. \quad (1.10)$$

Here and below the notation $\delta \in \Delta/\Lambda_{\hbar}$ refers to those chosen representatives. Thus $\mathcal{R}_{\Delta}$ represents $(\Delta + \Lambda_{\hbar})/(2\Lambda_{\hbar})$, with four representatives per original coset. Put

$$J_{b,\rho}(x) = L_{1,b}(x + \rho + \pi i\hbar(b - 1)), \quad b \geq 1, \quad \rho \in \mathcal{R}_{\Delta}, \quad D = \frac{d}{dx}. \quad (1.11)$$

Theorem 1.3 (Explicit nonlinear coefficient field). *Fix $\hbar > 0$ irrational and finite nonempty Δ . The functions $U = e^x$ and $V = e^{x/\hbar}$ are algebraically independent over $\mathbb{C}(x)$, and*

$$K_{\hbar,\Delta} = \mathbb{C}(x, U, V)(D^j J_{b,\rho} : b \geq 1, j \geq 0, \rho \in \mathcal{R}_{\Delta}). \quad (1.12)$$

All the displayed derivatives are algebraically independent over $\mathbb{C}(x, U, V)$. In particular the $J_{b,\rho}$ form a differential transcendence basis and $K_{\hbar,\Delta}$ is a purely differentially transcendental extension of $\mathbb{C}(x, U, V)$.

Let $\mathcal{A}_{\Delta}$ be an ordered complex basis of $\mathcal{V}_{\hbar,\Delta}$, and write $\mathcal{L}_{\Delta}$ for its Lyndon words. Introduce independent indeterminates $u, v, y_{b,\rho,j}$ indexed as in (1.12), and t_{ℓ} for $\ell \in \mathcal{L}_{\Delta}$. Every polynomial or rational expression uses only finitely many of them.

Corollary 1.4 (Absolute fixed-parameter functional completeness). *Evaluation gives a $\mathbb{C}(x)$ -algebra isomorphism*

$$\mathbb{C}(x)(u, v, y_{b,\rho,j})[t_{\ell} : \ell \in \mathcal{L}_{\Delta}] \xrightarrow{\sim} \mathcal{Q}_{\hbar,\Delta}^{>0}, \quad (1.13)$$

where $u \mapsto e^x$, $v \mapsto e^{x/\hbar}$, $y_{b,\rho,j} \mapsto D^j J_{b,\rho}$, and $t_{\ell} \mapsto I[\ell]$. Consequently these functions are jointly algebraically independent over $\mathbb{C}(x)$, and every element on the right reduces to a polynomial in the Lyndon integrals with rational coefficients in the displayed independent weight-zero variables.

The target in (1.13) includes the entire differential field $K_{\hbar,\Delta}$ by its definition in Theorem 1.2; it is not merely the positive-weight algebra over $\mathbb{C}(x)$. Thus Corollary 1.4 presents the nonlinear coefficient field as well as its shuffle extension. Theorem 1.3 is proved in Section 5; the combined statement is proved in Section 8.2.

1.4 What the proof establishes

The scalar proof has four steps. First, the two tails of (1.1) give a residue array with generating function $-d_X^{-a}d_Y^{-b}$. Irrationality makes its lattice labels injective. Second, the cumulative denominators in (1.2) are Laplace transformed into powers of gaps in an ordered simplex. Polynomial gap factors reduce to translated weight-zero letters by a Fourier integration-by-parts identity. Third, a minimal-word-length argument gives independence of Chen words over $K_{h,\Delta}$. Fourth, the two modular translations reduce the coefficient field to normalized mixed towers. Additive algebraic dependence would force a rational difference telescoper; a pole at a single position in an infinite multiplicative orbit excludes every such telescoper. This proves the coefficient-field presentation and removes the unspecified base field from the combined completeness statement.

Section 9 records the limits of these statements. In particular, continuation of a proved identity along a common path is not an independence theorem after numerical specialization. Radchenko–Wheeler’s algebraicity results for arithmetic special values [7] are related motivation, not an input to the proofs below. Their finite-scheme questions are not considered here.

2 Contours and meromorphic Fourier letters

Fix $\hbar > 0$ and write

$$\eta_{\hbar} = \min(1, \hbar^{-1}), \quad c_{a,b} = \pi(a + b\hbar).$$

For $0 < \eta < \eta_{\hbar}$ let $C_{\eta} = \mathbb{R} + i\eta$, oriented from left to right. No zero of either hyperbolic-sine factor lies in $0 < \Im p < \eta_{\hbar}$.

Lemma 2.1 (Fixed-height definition of the upper prescription). *For $|\Im z| < c_{a,b}$, the integral*

$$-i \int_{C_{\eta}} e^{-ipz} H_{a,b}(p) dp \tag{2.1}$$

is absolutely convergent, is independent of $\eta \in (0, \eta_{\hbar})$, and is holomorphic in z . It equals the contour with a small upper detour around $p = 0$. Thus it defines (1.1) without taking a nonuniform integrand limit at the origin.

Proof. On each fixed horizontal line avoiding the zeros of the denominator,

$$|H_{a,b}(t + i\eta)| \leq C_{\eta} e^{-c_{a,b}|t|}.$$

If $z = u + iv$, then $|e^{-i(t+i\eta)z}| = e^{tv+\eta u}$. On compact substrips the product is bounded by an integrable exponential, locally uniformly in z . The same remains true after any number of z -derivatives, since only powers of p are introduced. Cauchy’s theorem on rectangles between two positive heights applies; the vertical sides tend to zero by this estimate. Deformation to a path that is real outside a small disk and passes above the origin crosses no pole. $\square$

2.1 A two-tail expansion

For $a \in \mathbb{Z}_{\geq 0}$ set

$$C_a(m) = \begin{cases} \binom{a+m-1}{m}, & a > 0, \\ 1, & a = 0, \ m = 0, \\ 0, & a = 0, \ m > 0, \end{cases} \quad m \in \mathbb{Z}_{\geq 0}. \tag{2.2}$$

For $p > 0$ one has

$$H_{a,b}(p) = \sum_{m,n \geq 0} C_a(m)C_b(n)e^{-A_{m,n}p}, \quad A_{m,n} = \pi(a + 2m + \hbar(b + 2n)). \quad (2.3)$$

Terms with zero coefficient are omitted; every remaining $A_{m,n}$ is positive.

Proposition 2.2 (Normally convergent meromorphic expansion). *Choose $0 < \rho < \eta_{\hbar}$ and let Γ_{ρ} be the upper semicircle from $-\rho$ to ρ . Define the entire function*

$$E_{\rho}(z) = -i \int_{\Gamma_{\rho}} e^{-ipz} H_{a,b}(p) dp.$$

Then $L_{a,b}$ extends to a single-valued meromorphic function on $\mathbb{C}$, and

$$L_{a,b}(z) = E_{\rho}(z) - i \sum_{m,n \geq 0} C_a(m)C_b(n) \left\{ \frac{e^{-\rho(A_{m,n}+iz)}}{A_{m,n}+iz} + (-1)^{a+b} \frac{e^{-\rho(A_{m,n}-iz)}}{A_{m,n}-iz} \right\}. \quad (2.4)$$

Both series converge normally on compact sets away from their displayed poles. If $\hbar$ is irrational, all nonzero terms within either tail have distinct poles, and

$$\text{Res}_{z=iA_{m,n}} L_{a,b}(z) dz = -C_a(m)C_b(n), \quad \text{Res}_{z=-iA_{m,n}} L_{a,b}(z) dz = (-1)^{a+b} C_a(m)C_b(n). \quad (2.5)$$

In particular every pole is simple.

Proof. Use Lemma 2.1 to split the contour into its two real tails and Γ_{ρ} . On $[\rho, \infty)$ the expansion (2.3) is absolutely convergent. In the defining strip it may be integrated term by term, giving the first series in (2.4). On the negative tail put $p = -q$ and use $H_{a,b}(-q) = (-1)^{a+b} H_{a,b}(q)$.

The coefficients $C_a(m)C_b(n)$ have at most polynomial growth, whereas $e^{-\rho A_{m,n}}$ decays exponentially in every active index. On a fixed compact set of z values, all sufficiently large denominators have size comparable to $A_{m,n}$. The finitely many other denominators are bounded away from zero when the displayed poles are excluded. This proves normal convergence. The compact contour contribution is entire by differentiation under its integral.

The possible pole sets are locally finite, since $\hbar > 0$ and a bound on $A_{m,n}$ bounds all active indices. Equality of two positive pole labels implies $(m - m') + \hbar(n - n') = 0$, hence equality of the labels for irrational $\hbar$. The residues follow directly from the linear denominators in (2.4); their exponential numerators are 1 at the poles. $\square$

Remark 2.3. Although $\mathbb{Z} + \hbar\mathbb{Z}$ is dense in $\mathbb{R}$ when $\hbar$ is real irrational, the *positive cone* appearing in one letter's pole set is locally finite. Only finite unions of translated cones are used in any argument. The geometric expansions in (2.3) are never applied across $p = 0$.

2.2 Lowering, multiplication, and reflection

Proposition 2.4 (Exact letter identities). *For $a, b \geq 0$, $a + b > 0$, the meromorphic letters satisfy*

$$L_{a,b}(z + i\pi) - L_{a,b}(z - i\pi) = L_{a-1,b}(z) \quad (a \geq 1), \quad (2.6)$$

$$L_{a,b}(z + i\pi\hbar) - L_{a,b}(z - i\pi\hbar) = L_{a,b-1}(z) \quad (b \geq 1), \quad (2.7)$$

where the terminal symbol on the right is defined to be $L_{0,0} = 0$. Furthermore,

$$\begin{aligned} zL_{a,b}(z) &= i\pi a(L_{a+1,b}(z + i\pi) + L_{a+1,b}(z - i\pi)) \\ &\quad + i\pi\hbar b(L_{a,b+1}(z + i\pi\hbar) + L_{a,b+1}(z - i\pi\hbar)). \end{aligned} \quad (2.8)$$

Finally,

$$L_{a,b}(z) - (-1)^{a+b} L_{a,b}(-z) = P_{a,b}(z), \quad P_{a,b}(z) = -2\pi \operatorname{Res}_{p=0}(e^{-ipz} H_{a,b}(p)), \quad (2.9)$$

where $P_{a,b}$ is a polynomial of degree $a + b - 1$.

Proof. If the lowered index is nonterminal, subtract the integrands on a common convergence strip. The multipliers are $e^{\pi p} - e^{-\pi p}$ and $e^{\pi \hbar p} - e^{-\pi \hbar p}$, respectively. This proves (2.6)–(2.7) on a nonempty strip, and then everywhere by meromorphic continuation. These are also the depth-zero cases of [4, Section 3.0.1].

The terminal cases require a separate check; they are not integrals of a constant Fourier multiplier. Closing a rectangular contour upwards for $\Re z < 0$ gives

$$L_{1,0}(z) = \sum_{k \geq 1} (-1)^k e^{kz} = -\frac{e^z}{1 + e^z}, \quad L_{0,1}(z) = \hbar^{-1} L_{1,0}(z/\hbar). \quad (2.10)$$

For the first identity the residues at $p = ik$, $k \geq 1$, are $(-1)^k e^{kz}/(2\pi)$; heights $k + 1/2$ make the top edge tend to zero. The second follows by rescaling p . The required terminal centered differences vanish by the respective $2\pi i$ and $2\pi i \hbar$ periodicities. This justifies $L_{0,0} = 0$ solely as a notation for the terminal *identity*.

For (2.8), integrate $\partial_p(e^{-ipz} H_{a,b}(p))$ along C_η . The endpoints vanish exponentially in the defining strip. Since

$$H'_{a,b}(p) = -\pi a(e^{\pi p} + e^{-\pi p})H_{a+1,b}(p) - \pi \hbar b(e^{\pi \hbar p} + e^{-\pi \hbar p})H_{a,b+1}(p),$$

and $\int e^{-ipz} H_{a,b}(p) dp = iL_{a,b}(z)$, the result is (2.8). Every shifted integral converges in the original strip.

For reflection, substitute $p \mapsto -p$ in the reflected integral. The upper contour becomes the lower contour, and the parity of $H_{a,b}$ is $(-1)^{a+b}$. The upper contour minus the lower one is clockwise around the origin, so multiplication by $-i$ gives the factor -2π in (2.9). The pole order at the origin is $a + b$, and the residue of its product with e^{-ipz} is a polynomial of the stated degree. $\square$

Remark 2.5. The reflection correction is a polynomial, hence its product with dz is exact over $\mathbb{C}(z)$. It must not be discarded as a zero *function*. Our residue normalization gives $P_{1,0} = -1$, consistently with (2.10). Compare the quantum Bernoulli reflection formulas in [4, Section 5].

3 Residue tails and the complete letter quotient

Assume from now until Section 6 that $\hbar > 0$ is irrational. Put $T_{r,s}f(z) = f(z - i\pi(r + s\hbar))$. Use the rings $\mathcal{R}, \mathcal{S}$ and elements d_X, d_Y of (1.5).

3.1 The algebraic presentation

Lemma 3.1 (Localization presentation). *Let M be the $\mathcal{R}$ -module generated by $e_{a,b}$, $a, b \geq 0$, with*

$$e_{0,0} = 0, \quad d_X e_{a,b} = e_{a-1,b} \ (a \geq 1), \quad d_Y e_{a,b} = e_{a,b-1} \ (b \geq 1). \quad (3.1)$$

Then

$$M \xrightarrow{\sim} \mathcal{S}/\mathcal{R}, \quad e_{a,b} \mapsto [d_X^{-a} d_Y^{-b}]. \quad (3.2)$$

Proof. The displayed map respects every relation and is surjective. For injectivity, write an arbitrary finite element as $\sum P_{a,b}e_{a,b}$ and choose A, B bounding its indices. The relations give

$$\sum P_{a,b}e_{a,b} = Qe_{A,B}, \quad Q = \sum P_{a,b}d_X^{A-a}d_Y^{B-b}.$$

If its image in $\mathcal{S}/\mathcal{R}$ vanishes, then $Q = d_X^A d_Y^B P$ for some $P \in \mathcal{R}$. Thus $Qe_{A,B} = Pe_{0,0} = 0$. $\square$

By Proposition 2.4, X, Y may act on actual letters by $T_{1,0}$ and $T_{0,1}$; (3.1) then gives a well-defined map from M to their constant-linear span. We prove that this analytic map is injective.

3.2 Tail injectivity

There is an injective expansion map

$$\mathcal{S} \hookrightarrow \mathbb{C}[[X, Y]][X^{-1}, Y^{-1}], \quad d_X^{-a}d_Y^{-b} \mapsto \sum_{m,n \geq 0} C_a(m)C_b(n)X^{a+2m}Y^{b+2n}. \quad (3.3)$$

Indeed $d_X = (1 - X^2)/X$, and similarly for Y ; denominators $1 - X^2$ and $1 - Y^2$ are units in the formal power-series ring. The expansion has finite support if and only if the original element lies in $\mathcal{R}$. Consequently

$$\mathcal{S}/\mathcal{R} \hookrightarrow \mathbb{C}[[X, Y]][X^{-1}, Y^{-1}]/\mathcal{R} \quad (3.4)$$

is injective.

Theorem 3.2 (Residue-tail detection). *For a finite expression*

$$\Omega(z) = \sum c_{a,b,r,s} T_{r,s} L_{a,b}(z) dz \quad (3.5)$$

set

$$F_\Omega(X, Y) = \sum c_{a,b,r,s} X^r Y^s d_X^{-a} d_Y^{-b} \in \mathcal{S}.$$

If all finite residues of Ω vanish, then $F_\Omega \in \mathcal{R}$. In particular, a nonzero class $[F_\Omega] \in \mathcal{S}/\mathcal{R}$ forces a nonzero finite residue of Ω .

Proof. For a translated letter the positive poles from Proposition 2.2 have labels

$$(k, \ell) = (r + a + 2m, s + b + 2n), \quad z = i\pi(k + \hbar\ell), \quad (3.6)$$

and residues $-C_a(m)C_b(n)$. Their generating series is therefore $-X^r Y^s d_X^{-a} d_Y^{-b}$.

The negative poles have labels $(r - a - 2m, s - b - 2n)$. A finite collection of positive supports lies in a finite union of coordinatewise lower-bounded quadrants, and a finite collection of negative supports lies in a finite union of coordinatewise upper-bounded quadrants. Their intersection is finite. Since $(k, \ell) \mapsto k + \hbar\ell$ is injective on $\mathbb{Z}^2$, positive and negative poles can therefore cancel at only finitely many labels.

If all actual residues vanish, the expansion of F_Ω has finite support. The injectivity statement (3.4) implies $F_\Omega \in \mathcal{R}$. $\square$

Corollary 3.3 (One-coset exactness). *The map*

$$\mathcal{S}/\mathcal{R} \longrightarrow \text{span}_{\mathbb{C}}\{T_{r,s}L_{a,b}\}, \quad [X^r Y^s d_X^{-a} d_Y^{-b}] \mapsto T_{r,s}L_{a,b}$$

is an isomorphism. A form in this span is the differential of a globally single-valued meromorphic function if and only if it is the zero form.

Proof. Surjectivity and well-definedness follow from Lemma 3.1 and Proposition 2.4. A zero analytic expression has zero residues, so Theorem 3.2 makes its formal class zero; this proves injectivity. If G is meromorphic, its Laurent derivative has no residue at any finite point. Apply Theorem 3.2 and then the same injectivity argument. The converse is immediate. $\square$

Remark 3.4. The quotient by $\mathcal{R}$ in (3.2) removes *finite-support residue tails*. It is not an identification of Laurent multipliers with contact distributions. The passage from a zero tail class to an actual zero function uses the exact lowering identities, including the separately verified terminal identities. Residues by themselves would not exclude a nonzero entire remainder.

3.3 Offsets and an explicit basis

The possible poles of letters in the δ -coset lie in $-\delta + \Lambda_h$. For inequivalent offsets these sets are disjoint. For equivalent offsets, their letters are identified by the corresponding Laurent monomial. Applying Corollary 3.3 in each disjoint coset proves Theorem 1.1.

There is also a simple vector-space basis for the quotient, so the statement does not depend on an unspecified cohomology calculation.

Proposition 3.5 (Partial-fraction basis). *A complex basis of $\mathcal{S}/\mathcal{R}$ is given by the classes of*

$$(X - \epsilon)^{-a} Y^n, \quad \epsilon \in \{1, -1\}, \quad a \geq 1, \quad n \in \mathbb{Z}, \quad (3.7)$$

$$X^m (Y - \epsilon)^{-b}, \quad \epsilon \in \{1, -1\}, \quad b \geq 1, \quad m \in \mathbb{Z}, \quad (3.8)$$

$$(X - \epsilon)^{-a} (Y - \epsilon')^{-b}, \quad \epsilon, \epsilon' \in \{1, -1\}, \quad a, b \geq 1. \quad (3.9)$$

Every basis class is represented by a finite constant-linear combination of translated Fourier letters.

Proof. Partial fractions in one variable give the direct sum

$$\mathbb{C}[X^{\pm 1}, (1 - X^2)^{-1}] = \mathbb{C}[X^{\pm 1}] \oplus \bigoplus_{\epsilon=\pm 1, a \geq 1} \mathbb{C}(X - \epsilon)^{-a}.$$

Tensor this decomposition with its Y -analogue and quotient out the Laurent–Laurent summand. This gives precisely (3.7)–(3.9) and their independence. Finally

$$\frac{1}{X - \epsilon} = -\frac{X + \epsilon}{Xd_X}$$

expresses every such basis element in the presentation of Lemma 3.1; the assertion about letters follows. $\square$

4 Left-half-plane normalization and polynomial closure

The residue results are global, whereas the iterated integrals need a compatible normalization at an infinite endpoint. We establish it before using any bar construction.

Lemma 4.1 (Decay of all fixed translates). *Let $\hbar > 0$, and fix a, b with $a + b > 0$. For every compact interval $J \subset \mathbb{R}$ and every $0 < \eta < \eta_h$,*

$$L_{a,b}(u + iv) = O(e^{\eta u}) \quad (u \rightarrow -\infty), \quad v \in J. \quad (4.1)$$

The estimate is locally uniform for every fixed finite family of translates. Each function in $\mathcal{W}_{h,\Delta}$ is holomorphic in a sufficiently far left half-plane and satisfies such an estimate there.

Proof. On a compact substrip of $|\Im z| < c_{a,b}$, Lemma 2.1 on the line C_η gives (4.1) directly. For the one-frequency kernels, (2.8) and (2.6) give

$$L_{a+1,0}(z) = \frac{z + i\pi(a-1)}{2\pi ia} L_{a,0}(z - i\pi) \quad (a \geq 1). \quad (4.2)$$

Together with (2.10), this expresses $L_{a,0}$ as a polynomial of degree $a-1$ times a shifted logistic function. It therefore satisfies (4.1) on any bounded imaginary interval for $\eta < 1$. Rescaling gives $L_{0,b}(z) = \hbar^{-1} L_{b,0}(z/\hbar)$ and the assertion for $a = 0$.

For $a, b > 0$, rewrite (2.6) as

$$L_{a,b}(z + 2\pi i) = L_{a,b}(z) + L_{a-1,b}(z + \pi i).$$

Since $c_{a,b} > \pi$, any fixed bounded imaginary interval can be covered by finitely many translates of compact substrips of the defining strip by $2\pi i\mathbb{Z}$. Iterating the displayed identity reduces its additional terms to the case with smaller a . Induction, starting at $a = 0$, proves the estimate. All poles lie on the imaginary axis by Proposition 2.2; after finitely many offsets a left half-plane avoids them all. $\square$

Proposition 4.2 (Polynomial multiples remain in the letter space). *For every finite offset set Δ one has*

$$\mathbb{C}[x]\mathcal{W}_{\hbar,\Delta} \subseteq \mathcal{W}_{\hbar,\Delta}. \quad (4.3)$$

More explicitly, any polynomial multiple of a letter in (1.4) is a finite constant-linear combination of letters in its same offset coset.

Proof. For $u = x + \delta - i\pi(r + s\hbar)$, write $x = u - \delta + i\pi(r + s\hbar)$. Formula (2.8) expresses $uL_{a,b}(u)$ in translated letters. Iteration proves the result for powers of x , and then for all polynomials. $\square$

Remark 4.3 (Reflected letters). Reflection need not preserve the decay normalization at $-\infty$. By (2.9), adjoining reflected letters, and if necessary replacing Δ by $\Delta \cup (-\Delta)$, enlarges the alphabet only by polynomials. Their differentials are exact. In fact $\mathcal{W}_{\hbar,\Delta} \cap \mathbb{C}[x] = 0$ by Lemma 4.1, and Theorem 1.1 implies that an element of $(\mathcal{W}_{\hbar,\Delta} + \mathbb{C}[x])dx$ is meromorphically exact precisely when its $\mathcal{W}$ -component is zero. Arbitrary reflected words at an infinite endpoint require boundary subtraction or a finite basepoint; no such regularization is implicit in our main theorem.

5 The nonlinear weight-zero field

We now present the coefficient field itself. The residue theorem of Section 3 detects constant-linear exactness, but does not by itself exclude polynomial relations between meromorphic functions. For that purpose we use an additive difference equation and a rational pole-orbit obstruction. The general relation between telescoping and algebraic independence is classical; see [6, 5]. The particular lemmas needed below are proved directly. No differential closure of the difference constants, or assertion that an analytic extension has no new difference constants, is required.

Throughout this section $\hbar > 0$ is irrational, Δ is finite and nonempty, and $D = d/dx$. Use the representative set $\mathcal{R}_\Delta$ and the towers $J_{b,\rho}$ of (1.11). Write

$$U = e^x, \quad V = e^{x/\hbar}, \quad F_\hbar = \mathbb{C}(x, U, V), \quad \mathfrak{q} = e^{2\pi i/\hbar}. \quad (5.1)$$

Here U, V denote functions; the proof below identifies them with independent rational-function variables over $\mathbb{C}(x)$. The parameter $\mathfrak{q}$ in this section is distinct from the quantum-torus parameter in Appendix A. Define commuting automorphisms on $\text{Mer}(\mathbb{C})$ by

$$(\sigma f)(x) = f(x + 2\pi i), \quad (\tau f)(x) = f(x + 2\pi i \hbar).$$

They commute with D and preserve $F_\hbar$.

5.1 The elementary field and its difference constants

Lemma 5.1 (Elementary independence and fixed field). *The functions U, V are algebraically independent over $\mathbb{C}(x)$. On their rational function field,*

$$\sigma(x) = x + 2\pi i, \quad \sigma(U) = U, \quad \sigma(V) = \mathfrak{q}V, \quad F_\hbar^\sigma = \mathbb{C}(U). \quad (5.2)$$

Furthermore,

$$D = \partial_x + U\partial_U + \hbar^{-1}V\partial_V \quad \text{on } F_\hbar. \quad (5.3)$$

Proof. After clearing denominators, a polynomial relation between U and V would be a finite identity

$$\sum_{m,n} p_{m,n}(x) e^{(m+n/\hbar)x} = 0, \quad p_{m,n} \in \mathbb{C}[x].$$

Distinct pairs (m, n) give distinct real rates because $\hbar$ is irrational. Divide by the exponential with largest rate and let real $x \rightarrow +\infty$. All terms but its polynomial coefficient tend to zero. A nonzero polynomial cannot tend to zero in this way. Removing that term and repeating proves independence.

Put $E = \mathbb{C}(U)$, $t = 2\pi i$, and regard $F_\hbar = E(x, V)$. The automorphism fixes E and sends (x, V) to $(x + t, \mathfrak{q}V)$. Suppose $0 \neq f = P/Q \in F_\hbar$ is fixed, where $P, Q \in E[x, V]$ are coprime. The identity $\sigma(P)Q = P\sigma(Q)$ implies that P divides $\sigma(P)$. Total degree is preserved, so $\sigma(P) = cP$ with $c \in E^\times$; then also $\sigma(Q) = cQ$.

Write $P = \sum_n p_n(x) V^n$. For every nonzero coefficient polynomial,

$$\mathfrak{q}^n p_n(x + t) = c p_n(x).$$

Leading coefficients in x give $c = \mathfrak{q}^n$. As $\mathfrak{q}$ is not a root of unity, only one n can occur. The remaining p_n is a periodic polynomial, hence belongs to E . Thus $P = aV^n$. The same argument gives $Q = bV^m$, and the common multiplier forces $n = m$. Hence $f = a/b \in E$. This proves (5.2); (5.3) is the chain rule. $\square$

Remark 5.2. The difference-constant field in (5.2) is $\mathbb{C}(U)$, not $\mathbb{C}$. In particular the telescoping coefficients below must be allowed to be arbitrary rational functions of U . By contrast, the constants for the ordinary derivative D in every field between $\mathbb{C}$ and $\text{Mer}(\mathbb{C})$ are $\mathbb{C}$. The two notions of constants must not be interchanged.

5.2 Additive dependence implies a rational telescoper

Lemma 5.3 (Additive-dependence obstruction). *Let $F \subseteq E$ be difference fields of characteristic zero, with the same notation σ for their compatible automorphisms. Suppose $y_1, \dots, y_m \in E$ satisfy*

$$\sigma(y_i) = y_i + b_i, \quad b_i \in F.$$

If the y_i are algebraically dependent over F , then there are $c_i \in F^\sigma$, not all zero, and $g \in F$ such that

$$\sum_i c_i b_i = \sigma(g) - g. \quad (5.4)$$

No hypothesis on E^σ is made.

Proof. Let $\mathfrak{a}$ be the nonzero kernel of evaluation $F[Z_1, \dots, Z_m] \rightarrow E$, $Z_i \mapsto y_i$. The semilinear affine operator

$$\mathcal{T}(P) = \sigma(P)(Z_1 + b_1, \dots, Z_m + b_m)$$

preserves $\mathfrak{a}$, since evaluation of $\mathcal{T}(P)$ is $\sigma(P(y_1, \dots, y_m))$. Choose a graded monomial order and a nonzero $P \in \mathfrak{a}$ whose leading monomial is least possible, with leading coefficient normalized to 1. Translation preserves its leading monomial and coefficient. Therefore $\mathcal{T}(P) - P$, if nonzero, would contradict minimality. Thus $\mathcal{T}(P) = P$.

Let $n = \deg_Z P \geq 1$. An appropriate formal partial derivative of total order $n - 1$ is a nonconstant affine-linear polynomial

$$Q(Z) = c_0 + \sum_i c_i Z_i.$$

Formal partial derivatives in the Z_i commute with $\mathcal{T}$, because $b_i \in F$. Hence $\mathcal{T}(Q) = Q$. Comparing coefficients gives $\sigma(c_i) = c_i$ and $\sigma(c_0) + \sum_i c_i b_i = c_0$. At least one c_i is nonzero, and $g = -c_0$ proves the claim. The derivative Q need not itself lie in $\mathfrak{a}$; its invariance is all that is used. $\square$

We use only the implication in this lemma. The existence of a telescoper need not imply algebraic dependence if the extension introduces new σ -constants. This is why no converse, or unchanged-constants assertion, is implicit here. Applying the lemma to finitely many derivatives of the y_i is legitimate whenever $D\sigma = \sigma D$.

5.3 A pole at a single orbit position cannot telescope

In this subsection, poles mean valuations of the rational function field $E(x)(V)$ along the linear polynomials $V - \alpha$, with $\alpha \in \mathbb{C}^\times$. They are algebraic poles in the independent variable V , not assertions about a topological ordering of poles of a meromorphic function of x .

Lemma 5.4 (Finite-string obstruction). *Let E be a field containing $\mathbb{C}$, let $t \in \mathbb{C}^\times$, and let $q \in \mathbb{C}^\times$ have infinite multiplicative order. On $E(x, V)$ let σ fix E and send (x, V) to $(x + t, qV)$. If $r = \sigma(g) - g$ with $g \in E(x, V)$, then, in each orbit $\{\alpha q^n : n \in \mathbb{Z}\}$, the set of finite V -poles of r is either empty or contains at least two positions.*

Proof. For a nonzero g , the set

$$S = \{n \in \mathbb{Z} : g \text{ has a pole along } V - \alpha q^n\}$$

is finite. The corresponding set for $\sigma(g)$ is $S - 1$. Indeed, substitution by $(x + t, qV)$ transports the local valuation at one position to the preceding position and sends every nonzero leading coefficient in $E(x)$ to a nonzero one. If S is empty, so is the pole set of $\sigma(g) - g$ on the orbit. Otherwise let $m = \min S$, $n = \max S$. At $m - 1$ only $\sigma(g)$ has a pole; at n only g has a pole. Both poles survive in the difference, and $m - 1 < n$. An irreducible divisor not in this orbit cannot map into it under σ , since the inverse image of $V - \alpha q^k$ is again a constant linear divisor in the same orbit. Thus other, possibly x -dependent denominator factors of g do not affect this argument. $\square$

5.4 Normalized mixed towers and their increments

For $b \geq 1$ and $\rho \in \mathcal{R}_\Delta$, put

$$a_\rho = \exp((\rho + \pi i)/\hbar), \quad \alpha_\rho = -a_\rho^{-1}, \quad R_\rho(V) = -\frac{a_\rho V}{1 + a_\rho V} = -\frac{V}{V - \alpha_\rho}, \quad (5.5)$$

$$A_{b,\rho}(x) = \frac{\prod_{\ell=1}^{b-1} (x + \rho + \pi i + \pi i \hbar (2\ell - 1))}{\hbar (2\pi i \hbar)^{b-1} (b-1)!}. \quad (5.6)$$

The empty product gives $A_{1,\rho} = 1/\hbar$.

Proposition 5.5 (Tower difference equations). *The towers satisfy*

$$\sigma J_{b,\rho} - J_{b,\rho} = A_{b,\rho}(x) R_\rho(V). \quad (5.7)$$

For $b \geq 1$ they also satisfy

$$\tau^{-1} J_{b,\rho} = J_{b,\rho} - J_{b-1,\rho}, \quad J_{0,\rho}(x) := L_{1,0}(x + \rho - \pi i \hbar) \in F_\hbar. \quad (5.8)$$

The poles α_ρ lie in pairwise distinct $\mathfrak{q}$ -orbits, and $\deg_x A_{b,\rho} = b - 1$.

Proof. The first lowering identity gives

$$\sigma J_{b,\rho} - J_{b,\rho} = L_{0,b}(x + \rho + \pi i + \pi i \hbar (b - 1)).$$

Iterating (4.2) gives the explicit formula

$$L_{b,0}(z) = \frac{\prod_{\ell=1}^{b-1} (z + \pi i (2\ell - b))}{(2\pi i)^{b-1} (b-1)!} L_{1,0}(z - \pi i (b - 1)). \quad (5.9)$$

Use $L_{0,b}(z) = \hbar^{-1} L_{b,0}(z/\hbar)$ and substitute the preceding argument. The polynomial factor is (5.6), and the remaining logistic factor is (5.5). This proves (5.7). The second lowering identity, centered at $x + \rho + \pi i \hbar (b - 2)$, gives (5.8), including $b = 1$.

Finally,

$$\frac{\alpha_\rho}{\alpha_{\rho'}} \in \mathfrak{q}^\mathbb{Z} \iff \rho - \rho' \in 2\pi i (\mathbb{Z} + \hbar \mathbb{Z}) = 2\Lambda_\hbar.$$

This follows by using the kernel $2\pi i \mathbb{Z}$ of the complex exponential. The representatives are distinct modulo $2\Lambda_\hbar$. The degree assertion is immediate from the displayed product. $\square$

5.5 Independence of every finite collection of tower derivatives

Proposition 5.6 (No rational telescoper for the tower increments). *If finitely many $c_{b,\rho,j}(U) \in \mathbb{C}(U)$ and a function $g \in F_\hbar$ satisfy*

$$\sum_{b,\rho,j} c_{b,\rho,j}(U) D^j (A_{b,\rho}(x) R_\rho(V)) = \sigma(g) - g, \quad (5.10)$$

then every $c_{b,\rho,j}$ is zero.

Proof. Fix an active ρ and let j be the largest derivative order with a nonzero coefficient for this ρ . At the valuation $V = \alpha_\rho$,

$$D^j(A_{b,\rho}R_\rho) = \frac{(-1)^{j+1}j!\alpha_\rho^{j+1}\hbar^{-j}A_{b,\rho}(x)}{(V - \alpha_\rho)^{j+1}} + \text{terms of pole order at most } j. \quad (5.11)$$

For $j = 0$ this follows from $R_\rho = -1 - \alpha_\rho/(V - \alpha_\rho)$. Inductively, only the $(V/\hbar)\partial_V$ part of D raises the pole order; its leading value at the divisor is $\alpha_\rho/\hbar$. Derivatives of the polynomial in x give lower pole orders.

The leading coefficient in the ρ -part of the left side of (5.10) is therefore a fixed nonzero scalar times

$$\sum_b c_{b,\rho,j}(U)A_{b,\rho}(x).$$

It is nonzero: the $A_{b,\rho}$ have different degrees in the independent variable x , so they are linearly independent over $\mathbb{C}(U)$. Hence that part has a pole at α_ρ , and nowhere else in its $\mathfrak{q}$ -orbit. Other ρ' contribute no pole on the same orbit by Proposition 5.5. Thus the entire left side has exactly one pole position on this orbit, contradicting Lemma 5.4 with $E = \mathbb{C}(U)$, $t = 2\pi i$, and $q = \mathfrak{q}$. $\square$

Corollary 5.7 (Differential independence of the mixed towers). *The functions $D^j J_{b,\rho}$, for $b \geq 1$, $j \geq 0$, and $\rho \in \mathcal{R}_\Delta$, are algebraically independent over $F_\hbar$. Equivalently the $J_{b,\rho}$ are differentially algebraically independent over that differential field.*

Proof. Apply Lemma 5.3 to any finite collection of these functions in the difference field $\text{Mer}(\mathbb{C})$. Their increments are $D^j(A_{b,\rho}R_\rho)$ because D and σ commute. By Lemma 5.1 the resulting coefficients would lie in $\mathbb{C}(U)$. Proposition 5.6 excludes the nontrivial telescoper, so the finite collection is independent. Algebraic or differential-algebraic dependence would involve only a finite collection. $\square$

5.6 Reduction of every weight-zero generator

Proposition 5.8 (Generation by the normalized towers). *One has*

$$K_{\hbar,\Delta} = F_\hbar(D^j J_{b,\rho} : b \geq 1, j \geq 0, \rho \in \mathcal{R}_\Delta). \quad (5.12)$$

Every original translated Fourier letter and every derivative has a finite rational expression in the displayed generators. Such a reduction uses only (2.10), the two lowering identities, (2.8), and their ordinary differential consequences.

Proof. Fix one $\delta \in \Delta$. For $f = L_{1,0}(x + \delta)$ and $g = L_{0,1}(x + \delta)$,

$$U = -e^{-\delta} \frac{f}{1+f}, \quad V = -e^{-\delta/\hbar} \frac{\hbar g}{1+\hbar g}.$$

These identities of meromorphic functions prove $F_\hbar \subset K_{\hbar,\Delta}$; the denominators are not the zero functions. Each $J_{b,\rho}$ is an allowed lattice translate, and so are its derivatives. Thus the right side of (5.12) is contained in the left side. Call the right side F_* . It is a differential field by construction.

All axis letters belong to $F_\hbar$, by (5.9) and rescaling. Every $L_{1,b}(x + \eta)$ with $\eta \in \Delta + \Lambda_\hbar$ is $\sigma^n \tau^m J_{b,\rho}$ for some $n, m \in \mathbb{Z}$ and $\rho \in \mathcal{R}_\Delta$. The b -dependent offset in the tower is subtracted before choosing its representative modulo $2\Lambda_\hbar$. Equations (5.7) and (5.8) show that F_* is preserved by both shifts and their inverses. More explicitly,

$$\sigma^{-1} J_{b,\rho} = J_{b,\rho} - \sigma^{-1}(A_{b,\rho}R_\rho), \quad \tau J_{b,\rho} = J_{b,\rho} + \tau J_{b-1,\rho}.$$

The last equation is finite by induction on b , starting with $J_{0,\rho} \in F_{\hbar}$. Differentiating all these identities treats the jet generators. Thus all translates of the $a = 1$ mixed letters lie in F_* .

For $a \geq 1$ and $b \geq 1$, combine (2.8) at $z + \pi i$ with (2.6) for $L_{a+1,b}$ to obtain

$$\begin{aligned} L_{a+1,b}(z) &= \frac{z - \pi i(a-1)}{2\pi i a} L_{a,b}(z + \pi i) \\ &\quad - \frac{\hbar b}{2a} (L_{a,b+1}(z + \pi i + \pi i \hbar) + L_{a,b+1}(z + \pi i - \pi i \hbar)). \end{aligned} \quad (5.13)$$

Induction on the first index reduces every mixed letter to the preceding case. Each step lowers that index, although the second index can increase; hence the reduction is finite. The new offsets are still in $\Delta + \Lambda_{\hbar}$. Since F_* is differential, all derivatives also belong to it. This proves the reverse inclusion. $\square$

Proof of Theorem 1.3. Proposition 5.8 proves generation. Lemma 5.1 and Corollary 5.7 prove that U, V and all tower derivatives are jointly algebraically independent over $\mathbb{C}(x)$. Consequently evaluation identifies the indicated rational function field with $K_{\hbar, \Delta}$. The differential structure is given by (5.3) and $D(D^j J_{b,\rho}) = D^{j+1} J_{b,\rho}$. $\square$

Example 5.9 (One offset and the two terminal corrections). For $\Delta = \{0\}$ one may take

$$\mathcal{R}_{\Delta} = \{0, \pi i, \pi i \hbar, \pi i(1 + \hbar)\}.$$

For every $b \geq 1$ these give four differentially independent towers; all towers and all their derivatives are independent jointly over $F_{\hbar}$. For each representative,

$$\sigma J_{1,\rho} - J_{1,\rho} = \hbar^{-1} R_{\rho}(V), \quad \tau J_{1,\rho} - J_{1,\rho} = L_{1,0}(x + \rho + \pi i \hbar).$$

Thus the towers are not themselves periodic. The elementary corrections in both directions are essential, especially for the case $b = 1$.

Remark 5.10 (Exact scope of the nonlinear presentation). The pole calculation is in $\mathbb{C}(U)(x)(V)$; it does not rely on $|q|$ being different from 1. Infinite order is sufficient. Irrationality is nevertheless indispensable. At $\hbar = 1$ the integral gives $J_{1,0} = L_{1,1} = L_{2,0} \in \mathbb{C}(x, e^x)$, so the independence statement is false. Nor is independence asserted after adjoining arbitrary single-valued meromorphic functions: those could include the $J_{b,\rho}$ themselves. The coefficient-field presentation is precisely over $F_{\hbar}$.

6 The Laplace–simplex and Chen identification

This section only requires $\hbar > 0$, not irrationality. Let $a_j, b_j \geq 0$, $a_j + b_j > 0$, and $n_j \geq 1$.

Definition 6.1 (Several-variable upper prescription). In (1.2), use $p_j \in C_{\eta_j}$ with $0 < \eta_j < \eta_{\hbar}$. The notation $(\mathbb{R} + i0)^m$ means the value obtained with positive heights, continued within that positive-height chamber towards zero. It is not the product of independent real-axis principal values.

Lemma 6.2. *On the product strip*

$$|\Im \omega_j| < c_{a_j, b_j} \quad (1 \leq j \leq m), \quad (6.1)$$

(1.2) is absolutely convergent on every fixed set of positive heights and is independent of those heights.

Proof. For $q_j = p_1 + \cdots + p_j$, $\Im q_j = \eta_1 + \cdots + \eta_j > 0$, so $|q_j|^{-n_j} \leq (\eta_1 + \cdots + \eta_j)^{-n_j}$. The remaining factors have product exponential decay in the real momentum variables by the proof of Lemma 2.1. Vary one height at a time. All cumulative denominators involving that variable retain positive imaginary part throughout the deformation, and the hyperbolic-sine factors have no poles in the intervening strip. Rectangle estimates prove independence. $\square$

Theorem 6.3 (Laplace–simplex identity). *For ω in (6.1),*

$$\mathcal{F}_{\mathbf{a}, \mathbf{b}, \mathbf{n}}^h(\omega) = \int_{s_1 > \cdots > s_m > 0} \prod_{j=1}^m L_{a_j, b_j}(\omega_j - s_j) \frac{(s_j - s_{j+1})^{n_j-1}}{(n_j - 1)!} ds_1 \cdots ds_m, \quad s_{m+1} = 0. \quad (6.2)$$

The integral on the right is absolutely convergent and locally uniformly so. Both sides determine the same continued germ wherever the left side is continued along a common path. In particular, the right side gives a holomorphic continuation to $\Re \omega_j < 0$ for all j .

Proof. Take fixed positive heights and put $q_j = p_1 + \cdots + p_j$. For $n \geq 1$ and $\Im q > 0$,

$$q^{-n} = \frac{(-i)^n}{(n-1)!} \int_0^\infty t^{n-1} e^{iqt} dt. \quad (6.3)$$

Insert this identity for each $q_j^{-n_j}$. On a compact substrip the absolute value of the resulting integrand is bounded by a constant times

$$\prod_{j=1}^m e^{-\epsilon_j |\Re p_j|} \prod_{j=1}^m t_j^{n_j-1} e^{-(\eta_1 + \cdots + \eta_j) t_j}, \quad \epsilon_j > 0. \quad (6.4)$$

This is integrable in every real momentum and every $t_j > 0$, so Fubini applies. Set $s_j = t_j + \cdots + t_m$. This has Jacobian 1 and transforms the positive orthant into the ordered simplex in (6.2). Moreover,

$$\sum_j q_j t_j = \sum_j p_j s_j, \quad i^{|\mathbf{n}| - m} (-i)^{|\mathbf{n}|} = i^{-m}.$$

The separate p_j -integrals are $iL_{a_j, b_j}(\omega_j - s_j)$ by Lemma 2.1. Their factors of i cancel the prefactor, proving the formula for the actual positive-height prescription.

For convergence of the right side, Lemma 2.1 on fixed C_η gives $|L_{a_j, b_j}(\omega_j - s_j)| \leq C e^{-\eta s_j}$ on compact substrips. This dominates every polynomial gap factor. To reach $\Re \omega_j < 0$ with arbitrary imaginary parts, use Lemma 4.1 for the large s_j and holomorphy of each letter in the left half-plane for bounded s_j . These estimates are locally uniform, giving a holomorphic integral on the product of left half-planes. It agrees with the original function on the nonempty overlap. Uniqueness of analytic continuation proves the final assertion about common paths; it does not assert that a fixed undeformed simplex represents every continued branch. $\square$

Corollary 6.4 (An ascending Chen presentation). *Write $\omega_j = x + \delta_j$ with fixed offsets, allowing lattice shifts to be absorbed in them. Put $\theta_j(t) = L_{a_j, b_j}(t + \delta_j) dt$ and $e = dt$. On a sufficiently far left half-plane,*

$$\mathcal{F}_{\mathbf{a}, \mathbf{b}, \mathbf{n}}^h(x + \delta_1, \dots, x + \delta_m) = I[\theta_1 | e^{n_1-1} | \theta_2 | e^{n_2-1} | \cdots | \theta_m | e^{n_m-1}](x). \quad (6.5)$$

Here e^k means k consecutive letters, and I uses ascending order from $-\infty$ to x . In particular, when all $n_j = 1$ the right side is the pure word $I[\theta_1 | \cdots | \theta_m]$.

Proof. Set $t_j = x - s_j$ in (6.2). Then $-\infty < t_1 < \dots < t_m < x$, and, with $t_{m+1} = x$, $s_j - s_{j+1} = t_{j+1} - t_j$. Finally

$$\frac{(v-u)^k}{k!} = \int_{u < u_1 < \dots < u_k < v} du_1 \dots du_k.$$

This inserts exactly the displayed blocks of e 's. For complex x all variables run along the same horizontal ray; the parametrization by the real s_j fixes the orientation without an additional sign. $\square$

Remark 6.5. The letters θ_j in (6.5) are quantum Fourier transforms, not rational algebraic differentials. Thus this presentation is not an expression of irrational-parameter quantum polylogarithms as ordinary mixed-Tate iterated integrals. It gives a direct transform-level explanation of the shuffle structure recorded in [4, Theorem 3.7].

Proposition 6.6 (Elimination of the polynomial gap blocks). *Every function on the left of (6.5) is a finite $\mathbb{C}[x]$ -linear combination of pure words with letters in $\mathcal{W}_{h,\Delta}$. Every pure word in this space is a finite constant-linear combination of quantum polylogarithms with all weights $n_j = 1$.*

Proof. Rather than integrate exact letters separately from an infinite endpoint, expand the polynomial

$$\prod_{j=1}^m \frac{(t_{j+1} - t_j)^{n_j-1}}{(n_j - 1)!}, \quad t_{m+1} = x,$$

as a finite sum of monomials $cx^{k_0} \prod_j t_j^{k_j}$. Each $t_j^{k_j} L_{a_j, b_j}(t_j + \delta_j)$ is a finite constant-linear combination of translated letters by Proposition 4.2. Substitution into the absolutely convergent simplex integral gives the first claim. For the converse use multilinearity and the case $n_1 = \dots = n_m = 1$ of Corollary 6.4. $\square$

7 The shuffle differential algebra and word independence

We give the algebraic argument in the generality needed here, distinguishing a chosen letter space from the whole de Rham quotient of the coefficient field. Let (K, D) be a differential field of characteristic zero with constant field C , and let $V \subset K$ be a C -vector subspace such that

$$V \cap DK = 0. \tag{7.1}$$

Equivalently, the map $V \rightarrow K/DK$ is injective.

Definition 7.1 (Shuffle differential algebra). Set $\mathcal{B}(K, V) = K \otimes_C \text{Sh}_C(V)$. After choosing a basis $\{f_\alpha\}_{\alpha \in \mathcal{A}}$ of V , write its word tensors as $[\alpha_1 | \dots | \alpha_n]$. Extend D by

$$D_{\mathcal{B}}(g[\alpha_1 | \dots | \alpha_n]) = (Dg)[\alpha_1 | \dots | \alpha_n] + g f_{\alpha_n}[\alpha_1 | \dots | \alpha_{n-1}], \tag{7.2}$$

with the second term omitted for the empty word. The associated two-term differential complex is $\mathcal{B} \xrightarrow{D_{\mathcal{B}}} \mathcal{B} dx$.

The formula is a derivation for shuffle: the last letter of a shuffled word comes from exactly one of its two factors. It defines the usual one-variable reduced-bar model for the selected alphabet.

The constant word factor $\text{Sh}_C(V)$ has its deconcatenation Hopf structure; $\mathcal{B}$ is a commutative differential K -algebra and a left comodule algebra over that Hopf algebra via

$$g[\alpha_1 | \cdots | \alpha_n] \mapsto \sum_{j=0}^n [\alpha_1 | \cdots | \alpha_j] \otimes g[\alpha_{j+1} | \cdots | \alpha_n]. \quad (7.3)$$

Here differentiation acts only in the second factor. We do not assert that (7.2) is a Hopf-algebra derivation for a coproduct differentiated in both factors. No canonical choice of representatives of all of K/DK is being made.

Theorem 7.2 (Independence of iterated words). *Suppose an extension differential field realizes functions I_w indexed by words in $\mathcal{A}$, with $I_\emptyset = 1$ and*

$$DI_{v\alpha} = f_\alpha I_v. \quad (7.4)$$

Assume the coefficient field K embeds in this extension and the f_α represent C -linearly independent classes in K/DK . Then the I_w are linearly independent over K . If they satisfy the shuffle identities, evaluation $\mathcal{B}(K, V) \rightarrow K[I_w]$ is an isomorphism of differential K -algebras.

Proof. Suppose a nonzero formal relation $\sum_w g_w I_w = 0$ exists, with finite support. Choose one minimizing the maximal word length n , then minimizing the number of its nonzero length- n coefficients. Length zero is impossible because K embeds. Divide by a chosen top coefficient so that it becomes 1.

Differentiate the relation. By (7.4), its top coefficients are Dg_w , and one of them is zero. If any top coefficient is nonzero, the result is a nonzero relation with fewer top words, a contradiction. Thus all top coefficients are constants $c_w \in C$. The differentiated relation now has length at most $n - 1$, so by minimality it is formally zero. For each prefix v of length $n - 1$ its coefficient is

$$Dg_v + \sum_{\alpha} c_{v\alpha} f_\alpha = 0.$$

The classes of the f_α are independent in K/DK , hence all $c_{v\alpha}$ vanish. This contradicts the choice of the top coefficient. The first assertion follows. Shuffle makes evaluation an algebra map, and word independence proves its injectivity; surjectivity is its definition. $\square$

Remark 7.3. This minimal-length proof is the unipotent iterated-integral counterpart of the familiar antiderivative-independence argument. It is given in full so that no unstated differential-Galois hypothesis is required. The bar and iterated-integral background is classical; see [2, 1]. In particular the argument does not use an arithmetic theorem about values.

Corollary 7.4 (Constants and Lyndon generators). *The constants of $(\mathcal{B}(K, V), D_{\mathcal{B}})$ are C . For any total order on a basis of V , its shuffle algebra is polynomial on the Lyndon words. Under Theorem 7.2, the corresponding iterated integrals are algebraically independent over K .*

Proof. If $D_{\mathcal{B}}F = 0$, its longest word coefficients are constants. The coefficients one level lower, followed by (7.1), make every positive-length top coefficient zero. Repeating leaves $F \in C$. The polynomial presentation of a characteristic-zero shuffle algebra is Radford's theorem [18]; its application to iterated-integral algebras is also recalled in [1, Section 6]. Any finite polynomial relation here involves a finite subalphabet, so the usual finite-alphabet theorem suffices even when V is infinite-dimensional. $\square$

8 Functional completeness and absolute normal forms

Fix irrational $\hbar > 0$ and finite Δ . Let $K = K_{\hbar, \Delta}$ and $V = \mathcal{W}_{\hbar, \Delta}$. By Theorem 1.1, $V \cap DK = 0$ and $V \cong \mathcal{V}_{\hbar, \Delta}$. Because $K \subset \text{Mer}(\mathbb{C})$ contains $\mathbb{C}$, its differential constants are exactly $\mathbb{C}$.

Choose a complex basis of V , for example the explicit one obtained from Proposition 3.5 in each offset coset. On a half-plane

$$\Re x < -1 - \max_{\delta \in \Delta} \Re \delta,$$

all letters are holomorphic. Any finite family decays exponentially along leftward horizontal rays by Lemma 4.1. Thus all finite words (1.8) converge locally uniformly and are holomorphic there. One may verify holomorphy and path independence by first starting at $-R$ and then sending $R \rightarrow \infty$; vertical segments at $\Re x = -R$ have vanishing contribution. Differentiating at the endpoint gives (7.4) with the ascending convention. The ordinary partition of the product of two simplices gives

$$I[u](x)I[v](x) = \sum_{w \in u \amalg v} I[w](x), \quad (8.1)$$

including multiplicities.

Proof of Theorem 1.2. Theorem 7.2 gives an injective differential algebra map

$$K \otimes_{\mathbb{C}} \text{Sh}(V) \longrightarrow K[I_w].$$

It is surjective by construction. By Proposition 6.6, $K[I_w]$ equals the algebra generated by all positive-integral-weight quantum polylogarithms on the specified common-translation lines. Substituting $V \cong \mathcal{V}_{\hbar, \Delta}$ proves (1.9). The statement about Lyndon words follows from Corollary 7.4. $\square$

8.1 The meaning of complete relations

The proof supplies a finite reduction procedure for any finite proposed relation. Transform each positive-weight generator by (6.2); expand its polynomial gaps and use (2.8) to remove polynomial multiples of letters. Combine offsets in the same lattice coset. Reduce each letter by the partial-fraction basis of Proposition 3.5, then reduce products by shuffle. The result is a finite expression

$$\sum_w g_w(x) I[w](x), \quad g_w \in K, \quad (8.2)$$

where the words are in the chosen basis of $\mathcal{V}_{\hbar, \Delta}$. The original expression vanishes if and only if every g_w is zero in K . By Proposition 5.8 and Theorem 1.3, each coefficient reduces further to a rational expression in $x, U, V, D^j J_{b, \rho}$. Clearing denominators reduces its vanishing to a polynomial identity in algebraically independent variables. This last step, absent from a merely relative presentation, is now justified by Theorem 1.3.

The reductions consist of constant-linearity, elementary axis evaluation, the two modular letter relations, Fourier integration by parts, ordinary differentiation and rational arithmetic, the Laplace–Chen conversion, and shuffle. They give a complete finite symbolic reduction for every finite expression in the stated generators. This is an algebraic statement over $\mathbb{C}$: the offsets, $\hbar$, the representative choices, and their complex constants are treated as exact data. It is not a numerical decision algorithm for equality of arbitrary complex inputs or for membership in $\Lambda_{\hbar}$.

Corollary 8.1 (Larger single-valued coefficient fields). *The word-independence assertion remains valid over any differential field K' with $K_{\hbar, \Delta} \subseteq K' \subseteq \text{Mer}(\mathbb{C})$.*

Proof. Theorem 1.1 excludes primitives in all of $\text{Mer}(\mathbb{C})$, and the constant field of K' is still $\mathbb{C}$. Apply Theorem 7.2. $\square$

8.2 Combining coefficient and shuffle independence

Proof of Corollary 1.4. The coefficient evaluation in Theorem 1.3 is an isomorphism onto $K_{h,\Delta}$. By Theorem 1.2 and Corollary 7.4, adjoining the Lyndon integrals is a polynomial extension over that field. Composing these two isomorphisms gives (1.13). In particular, a polynomial relation among the combined generators can be regarded as a polynomial in the Lyndon variables with coefficients in $K_{h,\Delta}$. Their independence makes every such coefficient zero; coefficient-field independence then makes every original coefficient polynomial zero. This proves joint independence. $\square$

The differential structure in this explicit presentation is also determined:

$$Dx = 1, \quad DU = U, \quad DV = V/\hbar, \quad D(D^j J_{b,\rho}) = D^{j+1} J_{b,\rho},$$

and differentiation of words is (7.4). The letter coefficient on its right has an explicit tower expression by Proposition 5.8; shuffle reduction then expresses the result in Lyndon generators. Hence derivatives of positive-weight generators introduce no unspecified field of coefficients either.

Corollary 8.2 (Positive weights without adjoining the coefficient field). *Let $A_{h,\Delta}^{>0}$ be the $\mathbb{C}(x)$ -algebra generated by the same positive-weight quantum polylogarithms, without adjoining $K_{h,\Delta}$. Then*

$$A_{h,\Delta}^{>0} \cong \mathbb{C}(x) \otimes_{\mathbb{C}} \text{Sh}(\mathcal{V}_{h,\Delta}).$$

Proof. Restrict the injective map in Theorem 1.2 to coefficients in $\mathbb{C}(x)$. Its image is the stated algebra by Proposition 6.6. In particular surjectivity uses only polynomial coefficients in x , not arbitrary elements of $K_{h,\Delta}$. $\square$

Remark 8.3. Corollary 8.2 already follows from the relative shuffle theorem. The stronger content of Corollary 1.4 is that nonlinear identities inside the adjoined weight-zero differential field have now been classified, not that the positive-weight coefficient field was simply made smaller.

8.3 Small examples and normalization checks

Example 8.4 (The classical one-frequency sector). The first letters are

$$\begin{aligned} L_{1,0}(x) &= -\frac{e^x}{1+e^x}, \\ L_{2,0}(x) &= \frac{x}{2\pi i} \frac{e^x}{1-e^x}, \\ L_{3,0}(x) &= \frac{x^2 + \pi^2}{2(2\pi i)^2} \frac{-e^x}{1+e^x}. \end{aligned} \tag{8.3}$$

They follow from (2.10) and (4.2) and agree with [4, Lemma 5.4]. In particular,

$$I[L_{1,0}](x) = -\log(1+e^x), \quad I[L_{1,0}|L_{1,0}](x) = \frac{1}{2} \log^2(1+e^x),$$

with the branches fixed at $-\infty$. The factor $1/2$ is the shuffle multiplicity, not an extra relation in the letter space.

Example 8.5 (Eliminating one additional weight). For $\omega = x + \delta$, Theorem 6.3 and (2.8) give

$$\begin{aligned}\mathcal{F}_{a,b,2}^{\hbar}(\omega) &= \omega \mathcal{F}_{a,b,1}^{\hbar}(\omega) \\ &\quad - i\pi a \left(\mathcal{F}_{a+1,b,1}^{\hbar}(\omega + i\pi) + \mathcal{F}_{a+1,b,1}^{\hbar}(\omega - i\pi) \right) \\ &\quad - i\pi \hbar b \left(\mathcal{F}_{a,b+1,1}^{\hbar}(\omega + i\pi \hbar) + \mathcal{F}_{a,b+1,1}^{\hbar}(\omega - i\pi \hbar) \right).\end{aligned}\tag{8.4}$$

The constants are fixed by decay at $-\infty$. This identity illustrates why the depth of a pure-word presentation need not equal Goncharov's weight: the kernel orders a, b can increase when a polynomial gap is removed.

Example 8.6 (The two-frequency letter). For irrational $\hbar$, $L_{1,1}(z) dz$ has residue -1 at every $z = i\pi(1 + 2m + \hbar(1 + 2n))$ and residue $+1$ at the corresponding negative point. Its reflection identity is

$$L_{1,1}(z) - L_{1,1}(-z) = \frac{iz}{2\pi\hbar}.$$

The nonzero residues exclude a single-valued meromorphic primitive, even if one enlarges the coefficient field far beyond $K_{\hbar,\Delta}$.

9 Scope, functoriality, and remaining problems

9.1 The letter quotient is not the whole differential cohomology

The injection

$$\mathcal{V}_{\hbar,\Delta} \hookrightarrow K_{\hbar,\Delta} dx/dK_{\hbar,\Delta}$$

is generally not surjective. Choose z_0 outside the finite union $\bigcup_{\delta \in \Delta} (-\delta + \Lambda_{\hbar})$. Then $dx/(x - z_0)$ belongs to $K_{\hbar,\Delta} dx$ and has a nonzero residue at z_0 , while every quantum letter is regular there. It cannot be cohomologous to any finite quantum-letter combination. Accordingly $\mathcal{B}(K, V)$ here is the shuffle extension generated by a selected cohomologically independent alphabet, not a universal primitive closure of K . We make no assertion that its first de Rham cohomology vanishes.

9.2 Functional variables and continuation

Both completeness statements fix all offsets and vary one common translation x . They apply to every such fixed offset configuration, after lattice cosets are identified. It does not supply a universal algebraic presentation with the offsets themselves as varying coefficient parameters. Similarly, meromorphic continuation of the identities preserves equality of continued germs, but need not preserve independence after evaluating x at a number. A numerical or arithmetic specialization theorem would require new input.

The nonlinear theorem concerns the differential field generated by the *depth-one* weight-zero letters; the iterated-integral theorem concerns positive integral weights at arbitrary depth. These statements do not claim an independent classification for arbitrary mixed positive, zero, and negative weight indices in the original multiple-momentum definition.

The Laplace–Chen identity holds for every positive real $\hbar$. The explicit residue quotient and both independence arguments use irrationality. When $\hbar = r/s \in \mathbb{Q}$, pole labels collide; the injectivity of $(k, \ell) \mapsto k + \hbar \ell$ fails. The rational-parameter theory should not be inferred by substituting into the irrational-parameter independence statement. For example, at $\hbar = 1$ the kernels themselves give $L_{1,1} = L_{2,0}$, whereas $d_X^{-1} d_Y^{-1} - d_X^{-2}$ is nonzero in $\mathcal{S}/\mathcal{R}$. Thus neither the

irrational letter presentation nor the differential tower independence can remain unchanged at rational parameter. Goncharov treats reduction to multiple polylogarithms at rational parameter in [4, Theorems 3.2–3.3].

9.3 Functoriality requires actual differential data

For a differential-field isomorphism $\phi : K \rightarrow K'$ carrying a specified letter space V onto V' and commuting with derivations, there is a letterwise differential-algebra isomorphism

$$\mathcal{B}(K, V) \longrightarrow \mathcal{B}(K', V'), \quad g[f_1 | \cdots | f_n] \longmapsto \phi(g)[\phi(f_1) | \cdots | \phi(f_n)].$$

This statement concerns the *actual representative forms*. If only cohomology classes are transported, representative changes by exact forms can produce lower-length integration-by-parts and endpoint terms. The associated graded action alone does not determine the normalized action on iterated integrals.

Moreover a quantum-cluster transformation acts on a noncommutative quantum torus, not automatically on the commutative differential field $K_{h,\Delta}$. The operator theorem below proves a faithful presentation of a specified operator algebra. It does not assert that conjugation by the pentagon intertwiner carries scalar multiplication by every Fourier letter to scalar multiplication by another such letter. Consequently no universal cross-chart scalar completeness theorem is deduced here from the operator presentation.

9.4 Arithmetic motivation

Radchenko–Wheeler construct finite quantum dilogarithms from special values of a modular extension of the quantum dilogarithm and prove algebraicity [7]. Their second version gives additional algebraicity arguments and explicitly distinguishes the small-order normalization exceptions. Kopp interprets real-quadratic Stark class invariants using the Shintani–Faddeev modular cocycle [12]. Appleby–Flammia–Kopp’s earlier work gives the conditional Stark/ghost-SIC and rank- r framework [11]. Bengtsson–McConnell give complementary exact and numerical evidence relating SIC overlap units to products of square roots of Stark units from ray class fields [14]. The later Appleby–Flammia–Kopp paper proves the all-admissible-rank twisted convolution identity and supplies the RW/Shintani–Faddeev dictionary [8, Theorem 1.3 and Proposition 2.1]. These finite identities and arithmetic applications are background, not conclusions of this paper.

The shared analytic object motivates asking how a functional relation theory interacts with these special values. A literal description as fibers would require a defined family or specialization functor and a proof of its kernel. Neither the scalar theorem above nor the operator theorem below supplies that arithmetic comparison. Our companion note [17] studies the raw finite equations, with explicit low-order certificates and effective degree bounds, and finite étaleness of the raw solution scheme through a dual-number realization and invariant scalar reconstruction. None of those assertions is an input to the present functional or operator proofs.

A Faithfulness of the pentagon crossed product

This appendix proves the operator statement independently of the scalar completeness theorem. Its analytic inputs are Goncharov’s invariant Schwartz space and covariance and pentagon theorems [3, Theorems 1.1, 2.3, 2.6]. No inverse difference operator will be applied to a vector.

A.1 The Ore-denominator lemma

Theorem A.1 (Independence of intertwiner powers). *Let A be a left Ore domain over $\mathbb{C}$, let $D = \text{Frac}(A)$ be its fraction division ring, and let $\sigma \in \text{Aut}(A)$ extend to D . Suppose σ^j is outer on D for $1 \leq j < m$. Let $\pi : A \hookrightarrow \text{End}_{\mathbb{C}}(V)$ be faithful, and suppose $U \in \text{GL}_{\mathbb{C}}(V)$ satisfies*

$$U\pi(a) = \pi(\sigma(a))U \quad (a \in A). \quad (\text{A.1})$$

No action of D on V is required. Then

$$\sum_{j=0}^{m-1} \pi(a_j)U^j = 0 \implies a_j = 0 \text{ for all } j. \quad (\text{A.2})$$

Proof. Choose a nontrivial relation $R = \sum_j \pi(a_j)U^j = 0$ with the fewest nonzero coefficients. Right multiplication by the inverse of the smallest occurring power of U arranges $a_0 \neq 0$, with exponents still between 0 and $m-1$. There are at least two nonzero coefficients, by faithfulness and invertibility of U .

For any $a \in A$, choose $s \in A \setminus \{0\}$ and $t \in A$ such that

$$sa_0aa_0^{-1} = t \quad \text{in } D, \quad \text{hence } sa_0a = ta_0.$$

This is permitted by the left-fraction description of D . The relation $\pi(s)R\pi(a) - \pi(t)R = 0$ has coefficient zero at U^0 and reads

$$\sum_j \pi(sa_j\sigma^j(a) - ta_j)U^j = 0.$$

If any of its coefficients is nonzero in A , it contradicts minimal support. Therefore $sa_j\sigma^j(a) = ta_j$ in A for every j . Only now, in the abstract division ring, cancel s and write $d_j = a_0^{-1}a_j$. The result is

$$d_j\sigma^j(a) = ad_j \quad (a \in A).$$

For an active $j > 0$, $d_j \neq 0$, and this makes $\sigma^j(a) = d_j^{-1}ad_j$ on A , hence on its fraction division ring. This contradicts outerness. No active $j > 0$ remains, contrary to the choice of a relation with at least two terms. $\square$

Remark A.2. The argument never cancels the operator $\pi(s)$, which could have a kernel. It first obtains an equality of coefficients in A , and only then uses inverses in D . This distinction is the reason analytic Ore localization is unnecessary.

A.2 The modular-double coefficient algebra

Fix real irrational $\hbar > 0$ and set $q = e^{\pi i \hbar}$, $q^\vee = e^{\pi i / \hbar}$. All coefficient algebras in this appendix are specialized over $\mathbb{C}$ at these values. Let T be the tensor product of the two quantum tori with generators $X^{\pm 1}, Y^{\pm 1}, (X^\vee)^{\pm 1}, (Y^\vee)^{\pm 1}$ and relations

$$XY = q^2 YX, \quad X^\vee Y^\vee = (q^\vee)^2 Y^\vee X^\vee, \quad (\text{A.3})$$

with the two pairs commuting. Ordered Laurent monomials are a basis.

Let $\mathcal{L}$ be the specialized coefficient algebra denoted L in [3, Section 2.2]. We use two explicit properties established there: $\mathcal{L}$ is preserved by the order-five automorphism γ and

$$P = \mathbb{C}\langle X^{-1}, Y, (X^\vee)^{-1}, Y^\vee \rangle \subseteq \mathcal{L} \subseteq T. \quad (\text{A.4})$$

The first canonical-basis cone consists, up to nonzero powers of q , of $X^a Y^b$ with $a \leq 0$, $b \geq 0$, and similarly in the dual factor; this proves the first inclusion. On the fraction division ring, the first factor of γ is

$$\gamma(X) = Y^{-1}, \quad \gamma(Y) = (1 + qY)X, \quad (\text{A.5})$$

and the dual factor has the analogous formula with $q^\vee$. Goncharov proves $\gamma^5 = 1$.

Lemma A.3. *P , $\mathcal{L}$, and T are left and right Ore domains with the same fraction division ring D .*

Proof. Products of ordered monomials differ from the corresponding ordered monomial only by a nonzero scalar; leading-term comparison shows that all three algebras are domains. Let $P_{\leq n}$ be the span of monomials in its four generators of total degree at most n . Then $\dim P_{\leq n} = \binom{n+4}{4}$. For nonzero $a, b \in P$ of degrees at most d , the subspaces $P_{\leq n}a$ and $P_{\leq n}b$ both have this dimension and lie in $P_{\leq n+d}$. For large n ,

$$2\binom{n+4}{4} > \binom{n+d+4}{4}.$$

Their intersection is nonzero, giving $sa = tb$ with nonzero $s, t \in P$. The right-sided argument is identical, so P is Ore.

Its fraction division ring contains the inverses of the four generators, hence contains T . For $a, b \in \mathcal{L}$, $a, b \neq 0$, write $ab^{-1} = s^{-1}t$ with $s, t \in P \setminus \{0\}$. Then $sa = tb$ is a common left multiple in $\mathcal{L}$; right fractions give the right Ore condition. The same reasoning applies to T , and all fraction division rings equal $\text{Frac}(P)$. $\square$

Lemma A.4 (Outerness). *Every nonidentity power γ^j , $1 \leq j \leq 4$, is outer on D .*

Proof. For a nonzero Laurent polynomial in T , let $\deg_X$ be its largest X -exponent. Regarding T as a skew Laurent-polynomial ring in X over a domain shows $\deg_X(AB) = \deg_X(A) + \deg_X(B)$. It extends to $D^\times$ by

$$\deg_X(A^{-1}B) = \deg_X(B) - \deg_X(A).$$

To check well-definedness take common left Ore denominators; their degrees cancel by additivity. Every inner automorphism preserves this degree. But $\deg_X(X) = 1$ and $\deg_X(\gamma(X)) = \deg_X(Y^{-1}) = 0$, so γ is outer. If γ^j were inner for $1 \leq j \leq 4$, choose $r > 0$ with $jr \equiv 1 \pmod{5}$; then $\gamma = (\gamma^j)^r$ would be inner. This is impossible. $\square$

A.3 The analytic representation

On Gaussian-polynomial test functions, the torus generators act by

$$\begin{aligned} \hat{X}f(x) &= f(x + 2\pi i\hbar), & \hat{Y}f(x) &= e^x f(x), \\ \hat{X}^\vee f(x) &= f(x + 2\pi i), & \hat{Y}^\vee f(x) &= e^{x/\hbar} f(x). \end{aligned} \quad (\text{A.6})$$

Goncharov constructs a common invariant space $S_{\mathcal{L}}$ containing these test functions, and an invertible quantum-dilogarithm intertwiner K on it, such that

$$K^{-1}\hat{A}K = \widehat{\gamma(A)} \quad (A \in \mathcal{L}), \quad K^5 = \kappa I \quad (\kappa \in \mathbb{C}^\times). \quad (\text{A.7})$$

Here a fixed nonzero scalar normalization of his intertwiner is allowed; κ is determined by it. We import the common-domain and covariance assertions from [3, Theorems 1.1, 2.3, 2.6].

Lemma A.5 (Faithful coefficients). *The representation of $\mathcal{L}$ on $S_{\mathcal{L}}$ is faithful.*

Proof. Apply a Laurent monomial $Y^b(Y^\vee)^d X^a(X^\vee)^c$ to $f_0(x) = e^{-x^2/2}$. Its value is

$$e^{2\pi^2(ah+c)^2} e^{-x^2/2} e^{(b+d/h-2\pi i(ah+c))x}. \quad (\text{A.8})$$

For irrational real $\hbar$, the displayed exponents are distinct for distinct integer quadruples (a, b, c, d) : compare imaginary parts first, then real parts. Finite families of exponentials with distinct exponents are linearly independent, as follows for example from a Vandermonde determinant of their derivatives at zero. Thus the torus action on the Gaussian test space is faithful, and so is the restriction to $\mathcal{L}$ on $S_{\mathcal{L}}$. $\square$

Theorem A.6 (Pentagon crossed-product presentation). *For irrational real $\hbar > 0$, the finite algebraic operator relations in the algebra generated by $\widehat{\mathcal{L}}$ and K, K^{-1} are exactly the coefficient relations, covariance, and $K^5 = \kappa I$. More precisely,*

$$\mathcal{L}[u; \gamma^{-1}]/(u^5 - \kappa) \xrightarrow{\sim} \text{Alg}_{\mathbb{C}}(\widehat{\mathcal{L}}, K, K^{-1}), \quad u \longmapsto K. \quad (\text{A.9})$$

In particular $\sum_{j=0}^4 \widehat{A}_j K^j = 0$ with $A_j \in \mathcal{L}$ implies $A_0 = \dots = A_4 = 0$.

Proof. Covariance in (A.7) is $K\widehat{A} = \widehat{\gamma^{-1}(A)}K$. Apply Theorem A.1 with $A = \mathcal{L}$, $U = K$, $m = 5$, and $\sigma = \gamma^{-1}$, using Lemmas A.3 to A.5. This proves independence of the five powers. Since $\sigma^5 = 1$ and κ is a nonzero scalar, $u^5 - \kappa$ is central in the skew-polynomial ring. Division by this monic polynomial gives the unique left-coefficient normal form $\sum_{j=0}^4 A_j u^j$. The independence just proved makes its analytic realization injective; surjectivity follows from the definition of the generated algebra and $u^{-1} = \kappa^{-1}u^4$. $\square$

Remark A.7 (Operator scope). Theorem A.6 concerns finite sums and products in the specified generated algebra. It does not classify identities in an algebra enlarged by the bare Fourier transform, arbitrary meromorphic multiplication operators, analytic closures, or parameter-changing intertwiners. In particular it does not identify the operator algebra with the commutative shuffle algebra of Theorem 1.2.

Research and computational disclosure

The manuscript was developed with AI-assisted literature retrieval, proof exploration, drafting, and internal checking. The author is responsible for its mathematical claims and references. Symbolic low-order normalization checks and numerical contour comparisons accompany the draft as supplementary code; they are not used as proofs. This version is a research draft and has not undergone independent external refereeing.

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